9. Implementation of Subnetting in CISCO packet Tracer.

Aim:

Implementation of subnetting in CISCO packet Tracer Simulator

procedure:

i) Classicals Ip subnetting is a technique that allows for more efficient use of Ip address by allowing for subnet masks networks.

ii) CREATING A NETWORK TOPOLOGY:

The first step in implementing classless Ip Subnetting is to create a network topology in packet Trace.

To create a network technique to add devices.

iii) Adding the Devices:

Once we have created our network topology, we can add devices to it.

iv) SUBNETTING:

To Subnet the network address of 192.168.1.0/24 to provide enoug space for at least 5 addresses each

The Ip addressing for the network show in topology

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b Router R1

i) GigabitEthernet 0/0 : 192.168.1.1.

ii) GigabitEthernet 0/1 : 192.168.2.1

iii) Switch S1 :

v) FastEthernet 0/1 : 192.168.1.0/27.

vi) PC1 : 192.168.1.11

vii) PC2 : 192.168.1.12

viii) PC3 : 192.168.1.13

ix) PC4 : 192.168.1.14.

x). PC5 : 192.168.1.15

xi). FastEthernet 0/~~0~~ 2 : 192.168.2.0/27

xii) PC5 : 192.168.4.15.

Configure the device.

```
#enable
#Configure terminal
#interface FastEthernet 0/0
#ip. address {Ip address} {Subnet mask}
# no shutdown
#exit.
interface FastEthernet 0/1
Ip address {Ip address} {Subnet mask}
no Shutdown
exit
```

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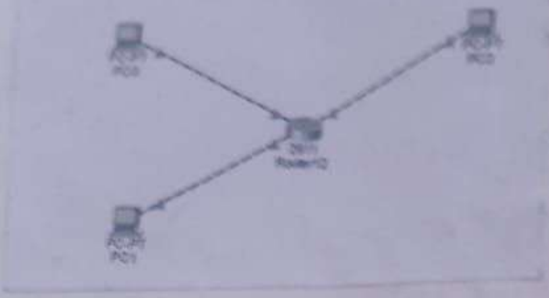
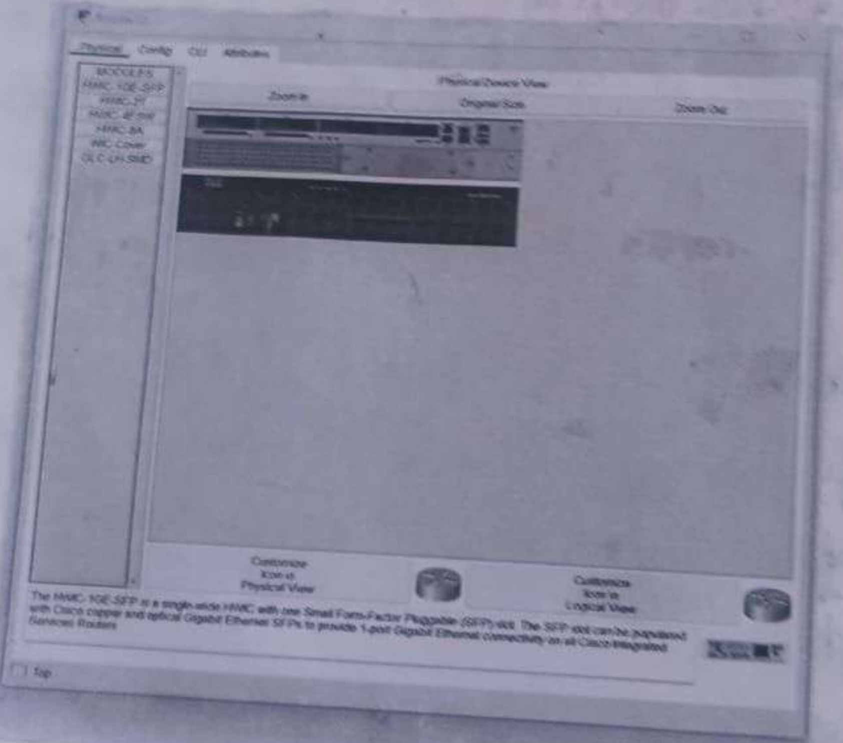
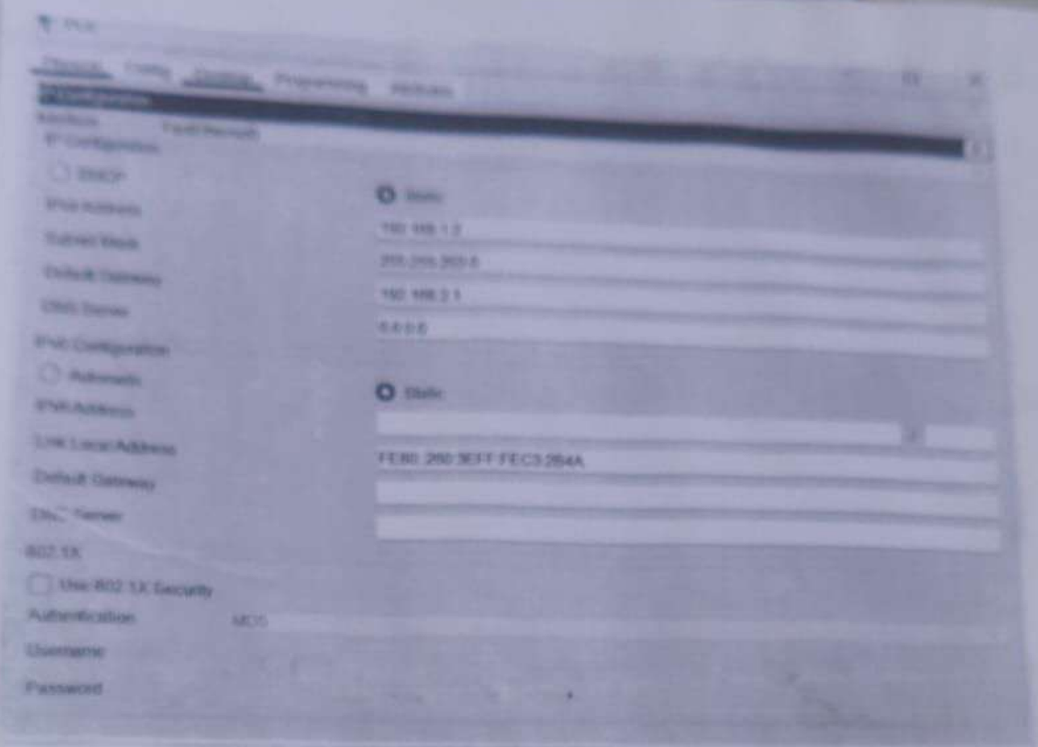




The HWIC-1GE-SFP is a single-wide HWIC with one Small Form-Factor Pluggable (SFP) slot. The SFP slot can be populated with Cisco copper and optical Gigabit Ethernet SFPs to provide 1-port Gigabit Ethernet connectivity on all Cisco Integrated Services Routers



IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

DNS Server: 0.0.0.0

Link Local Address: FE80::260:3EFF:FEC3:2B4A

Result:

Therefore, the Subnetting in CISCO packet Tracer Simulator was executed Successfully.

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